

Chapter 4. Vector Spaces (1/2)

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4.1. Vector Spaces and Subspaces

Vector spaces

- **Definition:** A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations, called *addition* and *multiplication by scalars* (real numbers), subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and d .
 1. The sum of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u} + \mathbf{v}$, is in V .
 2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
 4. There is a zero vector $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$
 5. For each $\mathbf{u}$ in V , there is a vector $-\mathbf{u}$ in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
 6. The scalar multiple of $\mathbf{u}$ by c , denoted by $c\mathbf{u}$, is in V .
 7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
 8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
 9. $c(d\mathbf{u}) = (cd)\mathbf{u}$
 10. $1\mathbf{u} = \mathbf{u}$

- Using these axioms, we can show that
 - the zero vector in Axiom 4 is unique, and
 - the vector $-\mathbf{u}$, called the **negative** of $\mathbf{u}$, in Axiom 5 is unique for each $\mathbf{u}$ in V .
 - *The identity and inverse with respect to vector addition are unique.*
- For each $\mathbf{u}$ in V and scalar c ,

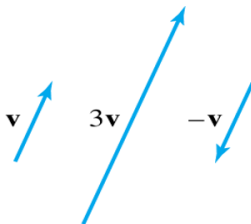
$$0\mathbf{u} = \mathbf{0}$$

$$c\mathbf{0} = \mathbf{0}$$

$$-\mathbf{u} = (-1)\mathbf{u}$$

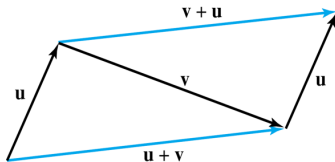
● Example 2:

- Let V be the set of all arrows (directed line segments) in three-dimensional space, with two arrows regarded as equal if they have the same length and point in the same direction.
- Define addition by the parallelogram rule, and for each $\mathbf{v}$ in V .
- Define $c\mathbf{v}$ to be the arrow whose length is $|c|$ times the length of $\mathbf{v}$, pointing in the same direction as $\mathbf{v}$ if $c \geq 0$ and otherwise pointing in the opposite direction.
- See the figure below. Show that V is a vector space.

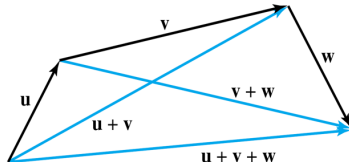


● Solution:

- The definition of V is geometric, using concepts of length and direction. No x y z -coordinate system is involved. An arrow of zero length is a single point and represents the zero vector.
- The negative of $\mathbf{v}$ is $(-1)\mathbf{v}$.
- So Axioms 1, 4, 5, 6, and 10 are evident. See the following figures.



$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$$



$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}).$$

Subspaces

- **Definition:** A **subspace** of a vector space V is a subset H of V that has three properties:
 - a. The zero vector of V is in H .
 - b. H is closed under vector addition. That is, for each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
 - c. H is closed under multiplication by scalars. That is, for each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .
- **Remark**
 - Properties (a), (b), and (c) guarantee that a subspace H of V is itself a vector space, under the vector space operations already defined in V .
 - Every subspace is a vector space.
 - Conversely, every vector space is a subspace (of itself and possibly of other larger spaces).

A subspace spanned by a set

- The set consisting of only the zero vector in a vector space V is a subspace of V , called the **zero subspace** and written as $\{0\}$.
- As the term **linear combination** refers to any sum of scalar multiples of vectors, and $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ denotes the set of all vectors that can be written as linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$.

- **Example 10:** Given $\mathbf{v}_1$ and $\mathbf{v}_2$ in a vector space V , let $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Show that H is a subspace of V .

- **Solution**

1. The zero vector is in H , since $0 = 0\mathbf{v}_1 + 0\mathbf{v}_2$.
2. To show that H is closed under vector addition, take two arbitrary vectors in H , say. $\mathbf{u} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2$ and $\mathbf{w} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2$. By Axioms 2, 3, and 8 for the vector space V ,

$$\mathbf{u} + \mathbf{w} = (s_1\mathbf{v}_1 + s_2\mathbf{v}_2) + (t_1\mathbf{v}_1 + t_2\mathbf{v}_2) = (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2$$

So, $\mathbf{u} + \mathbf{w}$ is in H .

3. Furthermore, if c is any scalar, then by Axioms 7 and 9,

$$c\mathbf{u} = c(s_1\mathbf{v}_1 + s_2\mathbf{v}_2) = (cs_1)\mathbf{v}_1 + (cs_2)\mathbf{v}_2$$

which shows that $c\mathbf{u}$ is in H and H is closed under scalar multiplication.

- Thus, H is a subspace of V .

- **Theorem 1:** If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in a vector space V , then $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is a subspace of V .
- We call $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ **the subspace spanned** (or **generated**) by $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
- Given any subspace H of V , a **spanning** (or **generating**) set for H is a set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in H such that $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

Suggested Exercise

- 4.1.13

4.2 Null spaces, Column spaces, and Linear transformation

Null space of matrix

- **Definition:** The **null space** of an $m \times n$ matrix A , written as $Nul A$, is the set of all solutions of the homogeneous equation $A\mathbf{x} = 0$. In set notation,

$$Nul A = \{\mathbf{x} : \mathbf{x} \in \mathbb{R}^n \text{ and } A\mathbf{x} = 0\}$$

- **Theorem 2:** The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$

- **Proof:**

- $Nul A$ is a subset of $\mathbb{R}^n$ because A has n columns.
- We need to show that $Nul A$ satisfies the three properties of a subspace.
 1. $\mathbf{0}$ is in $Nul A$ (trivial solution)
 2. Next, let $\mathbf{u}$ and $\mathbf{v}$ represent any two vectors in $Nul A$. Then, $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$. To show that $\mathbf{u} + \mathbf{v}$ is in $Nul A$, we must show that $A(\mathbf{u} + \mathbf{v}) = \mathbf{0}$. Using a property of matrix multiplication, we have $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$. Thus, $\mathbf{u} + \mathbf{v}$ is in $Nul A$, and $Nul A$ is closed under vector addition.
 3. Finally, if c is any scalar, then $A(c\mathbf{u}) = c(A\mathbf{u}) = c(\mathbf{0}) = \mathbf{0}$, which shows that $c\mathbf{u}$ is in $Nul A$.
- Thus, $Nul A$ is a subspace of $\mathbb{R}^n$

● An Explicit Description of $Nul A$

- There is no obvious relation between vectors in $Nul A$ and the entries in A .
- We say that $Nul A$ is defined *implicitly*, because it is defined by a condition that must be checked.
- No explicit list or description of the elements in $Nul A$ is given.
- *Solving* the equation $A\mathbf{x} = \mathbf{0}$ amounts to producing an *explicit* description of $Nul A$

$$A = \begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$$

1. The first step is to find the general solution of $A\mathbf{x} = 0$ in terms of free variables. Row reduce the augmented matrix $[A \mid 0]$ to *reduce* echelon form in order to write the basic variables in terms of the free variables:

$$A = \begin{pmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{rcl} x_1 - 2x_2 - x_4 + 3x_5 & = & 0 \\ x_3 + 2x_4 - 2x_5 & = & 0 \\ 0 & = & 0 \end{array}$$

2. The general solution is

- $x_1 = 2x_2 + x_4 - 3x_5$
- $x_3 = -2x_4 + 2x_5$
- x_2, x_4, x_5 free.

3. Next, decompose the vector giving the general solution into a linear combination of vectors where the weights are the free variables. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \\ = x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$$

4. Every linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is an element of $Nul A$. Thus, $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a spanning set for $Nul A$. $\square$

● Remark

1. The spanning set produced by the method in **Example 3** is automatically linearly independent because the free variables are the weights on the spanning vectors.
2. When $Nul A$ contains nonzero vectors, the number of vectors in the spanning set for $Nul A$ equals the number of free variables in the equation $A\mathbf{x} = \mathbf{0}$.

Column space of matrix

- **Definition:** The **column space** of an $m \times n$ matrix A , written as $Col A$, is the set of all linear combinations of the columns of A . If $A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$, then

$$Col A = Span\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$$

- **Theorem 3:** The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$

- **Remark**

- A typical vector in $Col A$ can be written as $A\mathbf{x}$ for some $\mathbf{x}$ because the notation $A\mathbf{x}$ stands for a linear combination of the columns of A . That is,

$$Col A = \{\mathbf{b} : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \text{ in } \mathbb{R}^n\}$$

- The notation $A\mathbf{x}$ for vectors in $Col A$ also shows that $Col A$ is the *range* of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$.
- The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ in $\mathbb{R}^m$.

● **Example 7:** Let

$$A = \begin{bmatrix} 2 & -4 & -2 & -1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$$

- Determine if $\mathbf{u}$ is in $\text{Nul } A$. Could $\mathbf{u}$ be in $\text{Col } A$?
- Determine if $\mathbf{v}$ is in $\text{Col } A$. Could $\mathbf{v}$ be in $\text{Nul } A$?

● **Solution to (a)**

- An explicit description of $\text{Nul } A$ is not needed here. Simply compute the product $A\mathbf{u}$.

$$A\mathbf{u} = \begin{bmatrix} 2 & -4 & -2 & -1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{u}$ is not a solution of $A\mathbf{x} = 0$, so $\mathbf{u}$ is not in $\text{Nul } A$. Also, with four entries, $\mathbf{u}$ could not possibly be in $\text{Col } A$, since $\text{Col } A$ is a subspace of $\mathbb{R}^3$.

• Solution to (b)

- Reduce $[A \mid \mathbf{v}]$ to an echelon form.

$$[A \mid \mathbf{v}] = \begin{bmatrix} 2 & -4 & -2 & -1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 & -2 & -1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

The equation $A\mathbf{x} = \mathbf{v}$ is consistent, so $\mathbf{v}$ is in $Col A$. With only three entries, $\mathbf{v}$ could not possibly be in $Nul A$, since $Nul A$ is a subspace of $\mathbb{R}^4$.

Kernel and range of linear transformation

- Subspaces of vector spaces other than $\mathbb{R}^n$ are often described in terms of a linear transformation instead of a matrix.
- **Definition:** A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each vector $\mathbf{x}$ in V a unique vector $T(\mathbf{x})$ in W , such that
 1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in V , and
 2. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}$ in V and all scalars c .
- **Definition:**
 - The **kernel** (or **null space**) of such a T is the set of all $\mathbf{u}$ in V such that $T(\mathbf{u}) = \mathbf{0}$ (the zero vector in W).
 - The **range** of T is the set of all vectors in W of the form $T(\mathbf{x})$ for some $\mathbf{x}$ in V .
- **Remark:**
 - The kernel of T is a subspace of V .
 - The range of T is a subspace of W .

Contrast between $Nul A$ and $Col A$ for an $m \times n$ matrix A

	$Nul A$	$Col A$
1	$Nul A$ is a subspace of $\mathbb{R}^n$	$Col A$ is a subspace of $\mathbb{R}^m$
2	$Nul A$ is implicitly defined, i.e., you are given only a condition ($A\mathbf{x} = 0$) that vectors in $Nul A$ must satisfy.	$Col A$ is explicitly defined, i.e., you are told how to build vectors in $Col A$
3	It takes time to find vectors in $Nul A$. Row operation on $[A \mid 0]$ are required.	It is easy to find vectors in $Col A$. The columns of A are displayed; others are formed from them.
4	There is no obvious relation between $Nul A$ and the entries in A .	There is an obvious relation between $Col A$ and the entries in A , since each column of A is in $Col A$.

(continued)

	<i>Nul A</i>	<i>Col A</i>
5	A typical vector $\mathbf{v}$ in <i>Nul A</i> has the property that $A\mathbf{v} = \mathbf{0}$.	A typical vector $\mathbf{v}$ in <i>Col A</i> has the property that the equation $A\mathbf{x} = \mathbf{v}$ is consistent.
6	Given a specific vector $\mathbf{v}$, it is easy to tell if $\mathbf{v}$ is in <i>Nul A</i> . Just compare $A\mathbf{v}$.	Given a specific vector $\mathbf{v}$, it may take time to tell if $\mathbf{v}$ is in <i>Col A</i> . Row operation on $[A \mid \mathbf{v}]$ are required.
7	$Nul A = \{\mathbf{0}\}$ if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.	$Col A = \mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^m$.
8	$Nul A = \{\mathbf{0}\}$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is <i>one-to-one</i> .	$Col A = \mathbb{R}^m$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ <i>onto</i> $\mathbb{R}^m$.

Suggested Exercises

- 4.2.5
- 4.2.17

4.3 Linearly independent sets; Bases

Linearly independent sets; Bases

- An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V is said to be **linearly independent** if the vector equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p = \mathbf{0} \quad (1)$$

has *only* the trivial solution, $c_1 = 0, \dots, c_p = 0$.

- The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if (1) has a nontrivial solution, *i.e.*, if there are some weights, $c_1, \dots, c_p$, *not all zero*, such that (1) holds. In such a case, (1) is called a **linear dependence relation** among $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
- **Theorem 4:** An indexed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.

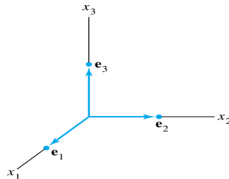
- **Definition:** Let H be a subspace of a vector space V . An indexed set of vectors $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ in V is a basis for H if
 - i) $\mathcal{B}$ is a linearly independent set, and
 - ii) The subspace spanned by $\mathcal{B}$ coincides with H ; that is, $H = \text{Span}\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$
- **Remark:**
 - The definition of a basis applies to the case when $H = V$, because any vector space is a subspace of itself. Thus, a basis of V is a linearly independent set that spans V .
 - When $H \neq V$, condition ii) includes the requirement that each of the vectors $\mathbf{b}_1, \dots, \mathbf{b}_p$ must belong to H , because $\text{Span}\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ contains $\mathbf{b}_1, \dots, \mathbf{b}_p$.

Standard basis

- Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the columns of the $n \times n$ matrix, I_n .
- That is,

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

- The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is called the **standard basis** for $\mathbb{R}^n$. See the following figure.



The standard basis for $\mathbb{R}^3$.

The spanning set theorem

- **Theorem 5:** Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a set in V , and let $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.
 - a. If one of the vectors in S — say, $\mathbf{v}_k$ — is a linear combination of the remaining vectors in S , then the set formed from S by removing $\mathbf{v}_k$ still spans H .
 - b. If $H \neq \{0\}$, some subset of S is a basis for H .

- **Proof for a.**

- By rearranging the list of vectors in S , if necessary, we may suppose that $\mathbf{v}_p$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_{p-1}$ — say,

$$\mathbf{v}_p = a_1 \mathbf{v}_1 + \dots + a_{p-1} \mathbf{v}_{p-1} \quad (3)$$

- Given any $\mathbf{x}$ in H , we may write

$$\mathbf{x} = c_1 \mathbf{v}_1 + \dots + c_{p-1} \mathbf{v}_{p-1} + c_p \mathbf{v}_p \quad (4)$$

for suitable scalars $c_1, c_2, \dots, c_p$.

- Substituting the expression for $\mathbf{v}_p$ from (3) into (4), it is easy to see that $\mathbf{x}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_{p-1}$.
- Thus, $\{\mathbf{v}_1, \dots, \mathbf{v}_{p-1}\}$ spans H , because $\mathbf{x}$ was an arbitrary element of H .

● Proof for b.

- If the original spanning set S is linearly independent, then it is already a basis for H .
- Otherwise, one of the vectors in S depends on the others and can be deleted, by part (a).
- So long as there are two or more vectors in the spanning set, we can repeat this process until the spanning set is linearly independent and hence is a basis for H .
- If the spanning set is eventually reduced to one vector, that vector will be nonzero (and hence linearly independent) because $H \neq \{0\}$.

● **Example 7:** Let

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}$$

and $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Note that $\mathbf{v}_3 = 5\mathbf{v}_1 + 3\mathbf{v}_2$, and show that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Then find a basis for the subspace H .

1. **Proof for $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$:**1.1 $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} \subset \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

- Every vector in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ belongs to H because $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + 0\mathbf{v}_3$.

1.2 $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subset \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$

- Now let $\mathbf{x}$ be any vector in H – say, $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$.
- Since $\mathbf{v}_3 = 5\mathbf{v}_1 + 3\mathbf{v}_2$, we may substitute

$$\begin{aligned} \mathbf{x} &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 \\ &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3(5\mathbf{v}_1 + 3\mathbf{v}_2) \end{aligned}$$

- Thus $\mathbf{x}$ is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, so every vector in H already belongs to $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ (1)

- We conclude that H and $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ are actually the same set of vectors.

2. Find a basis for the subspace H :

- It follows that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of H since $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.

Basis for $\text{Col } B$

- **Example 8:** Find a basis for $\text{Col } B$, where

$$B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Solution**

- Each non-pivot column of B is a linear combination of the pivot columns. In fact, $\mathbf{b}_2 = 4\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_2 - \mathbf{b}_3$. By the Spanning Set Theorem, we may discard $\mathbf{b}_2$ and $\mathbf{b}_4$, and $\{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\}$ will still span $\text{Col } B$.
- Let

$$S = \{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- Since $\mathbf{b}_1 \neq 0$ and no vector in S is a linear combination of the vectors that precede it, S is linearly independent. (Theorem 4). Thus, S is a basis for $\text{Col } B$.

Bases for $Nul A$ and $Col A$

- **Theorem 6:** The pivot columns of a matrix A form a basis for $Col A$.
- **Proof:**
 - Let B be the reduced echelon form of A . The set of pivot columns of B is linearly independent, for no vector in the set is a linear combination of the vectors that precede it.
 - Since A is row equivalent to B , the pivot columns of A are linearly independent as well, because any linear dependence relation among the columns of A corresponds to a linear dependence relation among the columns of B .
 - For this reason, every non-pivot column of A is a linear combination of the pivot columns of A .
 - Thus the non-pivot columns of A may be discarded from the spanning set for $Col A$, by the Spanning Set Theorem.
 - This leaves the pivot columns of A as a basis for $Col A$.
- **Warning:** The pivot columns of a matrix A are evident when A has been reduced only to echelon form. But, be careful to use the pivot columns of A itself for the basis of $Col A$. Row operations can change the column space of a matrix. The columns of an echelon form B of A are often not in the column space of A .

Two Views of a Basis

- When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent.
- If an additional vector is deleted, it will not be a linear combination of the remaining vectors, and hence the smaller set will no longer span V .
- Thus a basis is a spanning set that is as small as possible.
- A basis is also a linearly independent set that is as large as possible.
- If S is a basis for V , and if S is enlarged by one vector — say, $\mathbf{w}$ — from V , then the new set cannot be linearly independent, because S spans V , and $\mathbf{w}$ is therefore a linear combination of the elements in S .
- Sim: **“Basis is small enough to be linearly independent, but basis is large enough to span the space.”**

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